

1. Ideal I_{col0} with calculation

$V_{row0} = 1V$, $V_{row1} = 0V$, $V_{col0} = 0V$

Current into column 0 is equal to sum of current in $r[0][0]$ and $r[1][0]$

$$I_{col0} = (V_{row0} - V_{col0})/R[0][0] + (V_{row1} - V_{col0})/R[1][0]$$

$$I_{col0} = 1 / 1k + 0 = 1mA$$

2. KCL solution for floating point voltages

$V_{row0} = 1V$, $V_{col0} = 0V$, $V_{row1} = x$ (floating), $V_{col1} = y$ (floating)

Use KCL to get equations for V_{row1} and V_{col1}

Current leaving V_{row1} through $R[1][0]$ and $R[1][1]$ must sum to zero

$$X / 2k + (x-y) / 1k = 0$$

$$X + 2(x-y) = 0$$

$$3x - 2y = 0$$

Current leaving V_{col1} through $R[0][1]$ and $R[1][1]$ must sum to 0

$$(y-1)/2k + (y-x)/1k = 0$$

$$y-1 + 2(y-x) = 0$$

$$3y-2x = 1$$

Solve equations

Equation 1 gives $x = 2/3y$

Equation 2: $3y - 2(2/3)y = 1$

$$9/3y - 4/3y = 1$$

$$3/5 = 0.6 = y$$

$$X = 2/3 * 3/5 = 6/15 = 0.4$$

$$V_{row1} = x = 0.4V$$

$$V_{col1} = y = 0.6V$$

3. Actual I_{col0} with sneak path current

Column 0 receives current through both $R[0][0]$ and $R[1][0]$

$$I_{col0} = (V_{row0} - V_{col0})/R[0][0] + (V_{row1} - V_{col0})/R[1][0]$$

$$I_{col0} = (1-0) / 1k + (0.4-0)/2k = 1 + 0.2 = 1.2 \text{ mA}$$

$$\text{Sneak current} = 0.2\text{mA}$$

4. Explanation

Column current in a resistive crossbar array can be found using KCL, where the output column current is equal to the sum of the current through each crossbar (V/R). The floating row/column cause additional current to flow into the crossbars / output column. This extra “sneak-path” results in an output different from the intended dot product output.